Lemma 4.1.4 Let (ρ, V) and (σ, W) be irreducible A-modules (A assoc. alg.). Assume that $\dim V$ and $\dim W$ are at most countable. Then $\dim \mathrm{Hom}_A(V,W) = \begin{cases} 1 & \text{iff } (\rho,V) \simeq (\sigma,W) \\ 0 & \text{otherwise.} \end{cases}$

Proof Choose $T \in \mathrm{Hom}_A(V,W)$, $T \neq 0$. Then T should be a bijection because:

Step 1 $\ker(T)$ & $\mathrm{Im}(T)$ are A-modules.

Step 2 It follows that $\ker(T) = V$ & $\mathrm{Im}(T) = W$.

Next we show that $\dim \mathrm{Hom}_A(V,W) \leq 1$.

Step 3 Fix $T_0 \in \mathrm{Hom}_A(V,W)$. For all $S \in \mathrm{Hom}_A(V,W)$ set $R_S := T_0^{-1} S$. There are two cases:

Case 1. $\exists \lambda \in \mathbb{C}$ such that $R_S - \lambda I$ is not a bijection. Then by Step 1 & Step 2 we have $R_S - \lambda I = 0$; hence $R_S = \lambda I$, so that $S = \lambda T_0$.

Case 2. $\forall \lambda \in \mathbb{C}$, R_S is a bijection. Fix $v \in V$, $v \neq 0$ and consider

$$\{(R_S - \lambda I)^{-1} v\}_{\lambda \in \mathbb{C}}.$$

Since $\mathbb{C}$ is uncountable, these vectors are linearly dependent, i.e.,

$(R = R_S)$ $\sum_{i=1}^{N} c_i (R - \lambda_i)^{-1} v = 0$ We can assume all c_i's are nonzero & λ_i's are distinct

Set $f(x) := \sum c_i \prod_{j \neq i} (x - \lambda_j) \in \mathbb{C}[x]$. Then $f(\lambda_i) = c_i \prod_{j \neq i} (\lambda_i - \lambda_j) \neq 0$.

It follows that $f(x) \neq 0$.

Fact $f(R) v = 0$.

In particular, $f(x)$ is not a nonzero constant. Now factor $f(x)$:

$$f(x) = c(x - \mu_1) \cdots (x - \mu_r), \quad c \neq 0, \ r \geq 1.$$

$$0 = f(R) v = c (R - \mu_1 I) \cdots (R - \mu_r I) v.$$

If $(R - \mu_r I) v = 0$ then we are done (see Case 1.)

In general we can choose smallest $i>0$ such that $\prod_{j\geq i}(R-\mu_j I)v \neq 0$.

Set $w := \prod_{j\geq i}(R-\mu_j I)v$. Then $(R-\mu_{i-1})v = 0$. Again we are back to Case 1. ▨

Theorem 4.1.5 (Jacobson Density Theorem) Let A be a unital subalgebra of $\mathrm{End}(V)$ for a vector space V. Assume that $\dim V$ is at most countable and V is an irreducible A-module. Then the following statement is true for all $n\geq 1$:

> Given linearly independent elements $v_1,\dots,v_n \in V$, and arbitrary elements $w_1,\dots,w_n \in V$, there exists $a\in A$ such that $a\cdot v_i = w_i$ for $1\leq i\leq n$.

Proof Induction on n.

$n=1$: Follows from irreducibility of V.

$n \Rightarrow n+1$: Fix $v_1,\dots,v_{n+1}\in V$ linearly independent; $w_1,\dots,w_{n+1}$ arbitrary.

Step 1 By induction hypothesis, $\exists a_0\in A : a_0 v_i = w_i \quad 1\leq i\leq n$.

Step 2 Set $B := \{x\in A : xv_1 = \cdots = xv_n = 0\}$. Then Bv_{n+1} is an A-submodule.

$$b\in B,\ a\in A \Rightarrow abv_1 = \cdots = abv_n = 0 \Rightarrow ab\in B.$$

Step 3 If $Bv_{n+1}\neq\{0\}$ then $Bv_{n+1} = V$ by irreducibility of V, and we can set $a = a_0 + b$ for $b\in B$ such that $bv_{n+1} = w_{n+1} - av_{n+1}$.

Step 4 Assume $Bv_{n+1} = \{0\}$. Set $W := \{(xv_1,\dots,xv_{n+1}) : x\in A\} \subseteq V^{\oplus(n+1)}$.

In the rest, we show that this leads to a contradiction.

Also set $U = \{(\underbrace{0,\dots,0}_{n\text{ times}},v) : v\in V\} \subset V^{\oplus n+1}$

By induction hypothesis, $V^{\oplus(n+1)} = W+U$.

Step 5 If $W\cap U \neq \{(0,\dots,0)\}$ then

$$\exists x\in A : (xv_1,\dots,xv_{n+1}) = (0,\dots,0,w) \quad \& \ w\neq 0.$$

But $xv_1 = \cdots = xv_n = 0$ implies $x\in B$, hence $w = xv_{n+1} = 0$ (since $Bv_{n+1}=\{0\}$) which is a contradiction. Consequently, $W\cap U = \{(0,\dots,0)\}$, i.e.,

$$V^{\oplus(n+1)} = W\oplus U \quad \text{as } A\text{-modules.}$$

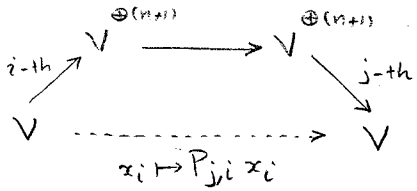
Step 6 Let $P: V^{\oplus(n+1)} \longrightarrow W \subseteq V^{\oplus(n+1)}$ be the projection $P(w,u) := w$.

Then $P \in \mathrm{Hom}_A(V^{\oplus(n+1)}, V^{\oplus(n+1)})$.

$$a(P(w,u)) = a(w) = a\cdot w \quad \& \quad P(a(w,u)) = P(aw,au) = aw.$$

Step 7 Write P in matrix form:

$$\forall x_1,\dots,x_{n+1} \in V: \quad P(x_1,\dots,x_{n+1}) = \Big(\sum_j P_{1,j}x_j,\ \sum_j P_{2,j}x_j,\ \dots,\ \sum_j P_{n+1,j}x_j\Big)$$



Therefore $P_{j,i} \in \mathrm{Hom}_A(V,V)$ for all $1 \le i,j \le n+1$.

Step 8 By Schur's Lemma, $P_{j,i} = c_{j,i} I_{V,V}$ for all $1 \le i,j \le n+1$. In particular, for every subspace $Z \subseteq V$ we have $P(Z^{\oplus(n+1)}) \subseteq Z^{\oplus n+1}$. Moreover, for every linear transformation $T: V \longrightarrow V$ we have $PT^{\oplus(n+1)} = T^{\oplus(n+1)}P$, where

$$T^{\oplus(n+1)}: V^{\oplus(n+1)} \longrightarrow V^{\oplus(n+1)}, \quad T^{\oplus(n+1)}(v_1,\dots,v_{n+1}) := (Tv_1,\dots,Tv_{n+1})$$

$$\begin{bmatrix} T & & \\ & T & \\ & & \ddots \end{bmatrix}\begin{bmatrix} c_{11} & \cdots & c_{1,n+1} \\ \vdots & & \end{bmatrix} = \begin{bmatrix} c_{11}T & \cdots & c_{1,n+1}T \\ \vdots & & \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1,n+1} \\ \vdots & & \end{bmatrix}\begin{bmatrix} T & & \\ & T & \\ & & \ddots \end{bmatrix}$$

Step 9 Set $Z := \mathrm{Span}\{v_1,\dots,v_{n+1}\}$. Choose arbitrary $\tilde{w}_1,\dots,\tilde{w}_{n+1} \in V$. Define a linear map

$T: Z \longrightarrow V$, $T(v_i) = \tilde{w}_i$, $1 \le i \le n+1$, and extend it to a linear map $T: V \longrightarrow V$.

$$(*) \qquad T^{(n+1)}P(v_1,\dots,v_{n+1}) = PT^{(n+1)}(v_1,\dots,v_{n+1}) = P(\tilde{w}_1,\dots,\tilde{w}_{n+1})$$

Also, $(v_1,\dots,v_{n+1}) = (1\cdot v_1,\dots,1\cdot v_{n+1}) \Rightarrow (v_1,\dots,v_{n+1}) \in W \Rightarrow P(v_1,\dots,v_{n+1}) = (v_1,\dots,v_{n+1})$

($1 \in A$)

$$\Rightarrow T^{(n+1)}P(v_1,\dots,v_{n+1}) = T^{(n+1)}(v_1,\dots,v_{n+1})$$

$(**)$

$$= (\tilde{w}_1,\dots,\tilde{w}_{n+1}).$$